Name:

1. Rationalizing denominators

Goal: Rewrite a fraction so there is no square root in the denominator. Multiplying the **top and bottom** by the same thing multiplies by 1, so the value does not change.

Simple square root in denominator: Multiply by that square root. $\dfrac{6}{\sqrt{3}} = \dfrac{6}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$

A sum or difference: Multiply by the **conjugate**, which flips the sign between the terms. It works because $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$, which has no square root.

$$\frac{8}{3+\sqrt{5}} = \frac{8}{3+\sqrt{5}} \cdot \frac{3-\sqrt{5}}{3-\sqrt{5}} = \frac{8(3-\sqrt{5})}{9-5} = 2(3-\sqrt{5}) = 6 - 2\sqrt{5}$$

Rationalize each denominator and simplify.

(a) $\dfrac{3}{\sqrt{6}}$

(b) $\dfrac{6}{\sqrt{3}-1}$

(c) $\dfrac{\sqrt{5}}{2+\sqrt{5}}$

2. Scaling a vector

Length: The vector $[a, b]$ has length $\sqrt{a^2 + b^2}$.

Length 1: Divide each component by the length. **Length k:** Multiply the length-1 vector by k. **Opposite direction:** Use $-k$.

Example: $[1, 1]$ has length $\sqrt{2}$. Same direction, length 1: $\left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$. Opposite direction, length 6: $\left[\frac{-6}{\sqrt{2}}, \frac{-6}{\sqrt{2}}\right] = \left[-3\sqrt{2}, -3\sqrt{2}\right]$.

Given the vector $[2, 1]$, find each vector. Rationalize any denominators.

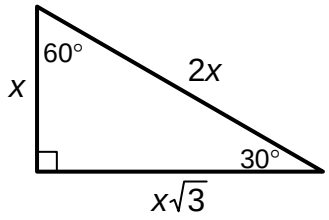
(a) Same direction, length 1

(b) Opposite direction, length 10

3. The 30-60-90 triangle

Ratio: In a 30-60-90 triangle, the sides are x (opposite 30°), $x\sqrt{3}$ (opposite 60°), and $2x$ (the hypotenuse).

Example: If the long leg is 6, then $x\sqrt{3} = 6$, so $x = \dfrac{6}{\sqrt{3}} = 2\sqrt{3}$, and the hypotenuse is $2x = 4\sqrt{3}$.



A 30-60-90 triangle has perimeter 12. Find the length of its shorter leg. Rationalize the denominator. *Hint: Write the perimeter in terms of x, then factor out x.*